

A Markov Chain Problem

Shengqi Qiu

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This problem below comes from one of the questions on the first midterm exam of MATH 632, taught by Professor Benedek Valko in Fall 2025. I am indebted to Prof. Valko in many ways, and I remember solving this problem on that midterm as one of the most amazing moments of my undergraduate years. Moreover, MATH 632 is the only course I ever attended from beginning to end, missing only one lecture. It changed a belief I had held for many years: that one could only learn mathematics through self-study. It showed me, perhaps for the first time, how much one can learn from listening carefully to someone who understands a subject deeply and knows how to explain it.

The Problem

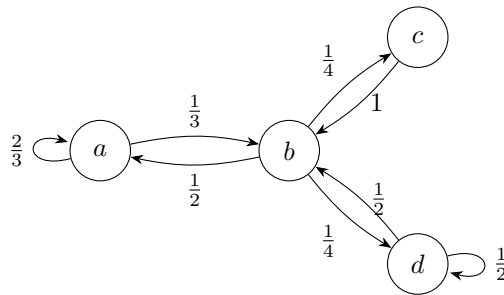
Consider the Markov chain on the state space

$$S = \{a, b, c, d\}$$

given by the transition diagram below. Find

$$\mathbb{P}_b(T_a < T_c),$$

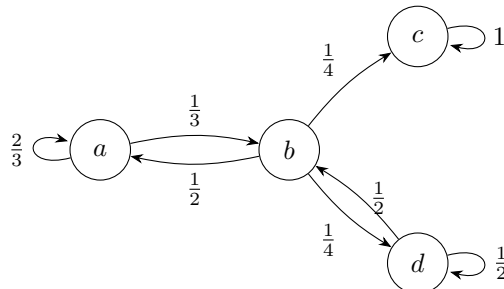
the probability that the Markov chain started from b visits a before c .



My Solution During the Exam.

It amounts to making c absorbing.

Hence the transition diagram becomes



We have

$$\begin{aligned}
\mathbb{P}_b(T_a < T_c) &= \mathbb{P}_b(T_a < T_c \mid X_1 = c) P(b, c) \\
&\quad + \mathbb{P}_b(T_a < T_c \mid X_1 = d) P(b, d) \\
&\quad + \mathbb{P}_b(T_a < T_c \mid X_1 = a) P(b, a) \\
&= 0 + P(b, a) + \frac{1}{4} \mathbb{P}_b(T_a < T_c \mid X_1 = d) \\
&= \frac{1}{2} + \frac{1}{4} \mathbb{P}_d(T_a < T_c).
\end{aligned}$$

Now,

$$\begin{aligned}
\mathbb{P}_d(T_a < T_c) &= \mathbb{P}_d(T_a < T_c \mid X_1 = d) P(d, d) \\
&\quad + \mathbb{P}_d(T_a < T_c \mid X_1 = b) P(d, b) \\
&= \frac{1}{2} \mathbb{P}_d(T_a < T_c) + \frac{1}{2} \mathbb{P}_b(T_a < T_c).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathbb{P}_b(T_a < T_c) &= \frac{1}{2} + \frac{1}{4} \left(\frac{1}{2} \mathbb{P}_d(T_a < T_c) + \frac{1}{2} \mathbb{P}_b(T_a < T_c) \right) \\
&= \frac{1}{2} + \frac{1}{4} \left[\frac{1}{2} \left(\frac{1}{2} \mathbb{P}_d(T_a < T_c) + \frac{1}{2} \mathbb{P}_b(T_a < T_c) \right) + \frac{1}{2} \mathbb{P}_b(T_a < T_c) \right].
\end{aligned}$$

Continuing this expansion,

$$\mathbb{P}_d(T_a < T_c) = \sum_{h=1}^n \frac{1}{2^h} \mathbb{P}_b(T_a < T_c) + \frac{1}{2^n} \mathbb{P}_d(T_a < T_c).$$

Let $n \rightarrow \infty$. Since

$$\frac{1}{2^n} \mathbb{P}_d(T_a < T_c) \rightarrow 0$$

and

$$\sum_{h=1}^{\infty} \frac{1}{2^h} = 1,$$

we obtain

$$\mathbb{P}_d(T_a < T_c) = \mathbb{P}_b(T_a < T_c).$$

Hence

$$\mathbb{P}_b(T_a < T_c) = \frac{1}{2} + \frac{1}{4} \mathbb{P}_b(T_a < T_c),$$

so

$$\frac{3}{4} \mathbb{P}_b(T_a < T_c) = \frac{1}{2}.$$

Therefore,

$$\boxed{\mathbb{P}_b(T_a < T_c) = \frac{2}{3}}.$$

Why The Problem Moved Me

What I found beautiful about this problem only became clear to me after the exam. During the exam, I did not remember the standard matrix method for absorbing Markov chains. I simply followed the probabilistic structure of the chain: starting from state d , the process either returns to b with probability $1/2$, or stays at d with probability $1/2$. This gives the recursion

$$p_d = \frac{1}{2}p_b + \frac{1}{2}p_d,$$

where $p_x = \mathbb{P}_x(T_a < T_c)$. Instead of immediately solving this linear equation algebraically, I repeatedly substituted the expression for p_d into itself:

$$p_d = \frac{1}{2}p_b + \frac{1}{2}\left(\frac{1}{2}p_b + \frac{1}{2}p_d\right) = \left(\frac{1}{2} + \frac{1}{2^2}\right)p_b + \frac{1}{2^2}p_d,$$

and, after n repetitions,

$$p_d = \left(\sum_{k=1}^n \frac{1}{2^k}\right)p_b + \frac{1}{2^n}p_d.$$

Since $0 \leq p_d \leq 1$, the remaining term satisfies

$$\frac{1}{2^n}p_d \longrightarrow 0,$$

while

$$\sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$$

Thus I strikingly found that

$$p_d = p_b.$$

After the exam, a classmate told me that the intended solution was the matrix formula

$$U = (I - Q)^{-1}S.$$

Only then did I realize what I had accidentally reconstructed. For a transient matrix Q ,

$$(I - Q)^{-1} = I + Q + Q^2 + Q^3 + \cdots,$$

which is precisely the **Neumann series**!!! The factor $2^{-n}p_d$ that I watched vanish was, in miniature, the same phenomenon as the transient contribution Q^n disappearing as $n \rightarrow \infty$. What moved me was that I had not consciously recalled the theorem at all. I had simply followed the structure of the problem far enough that the theorem re-emerged on its own!